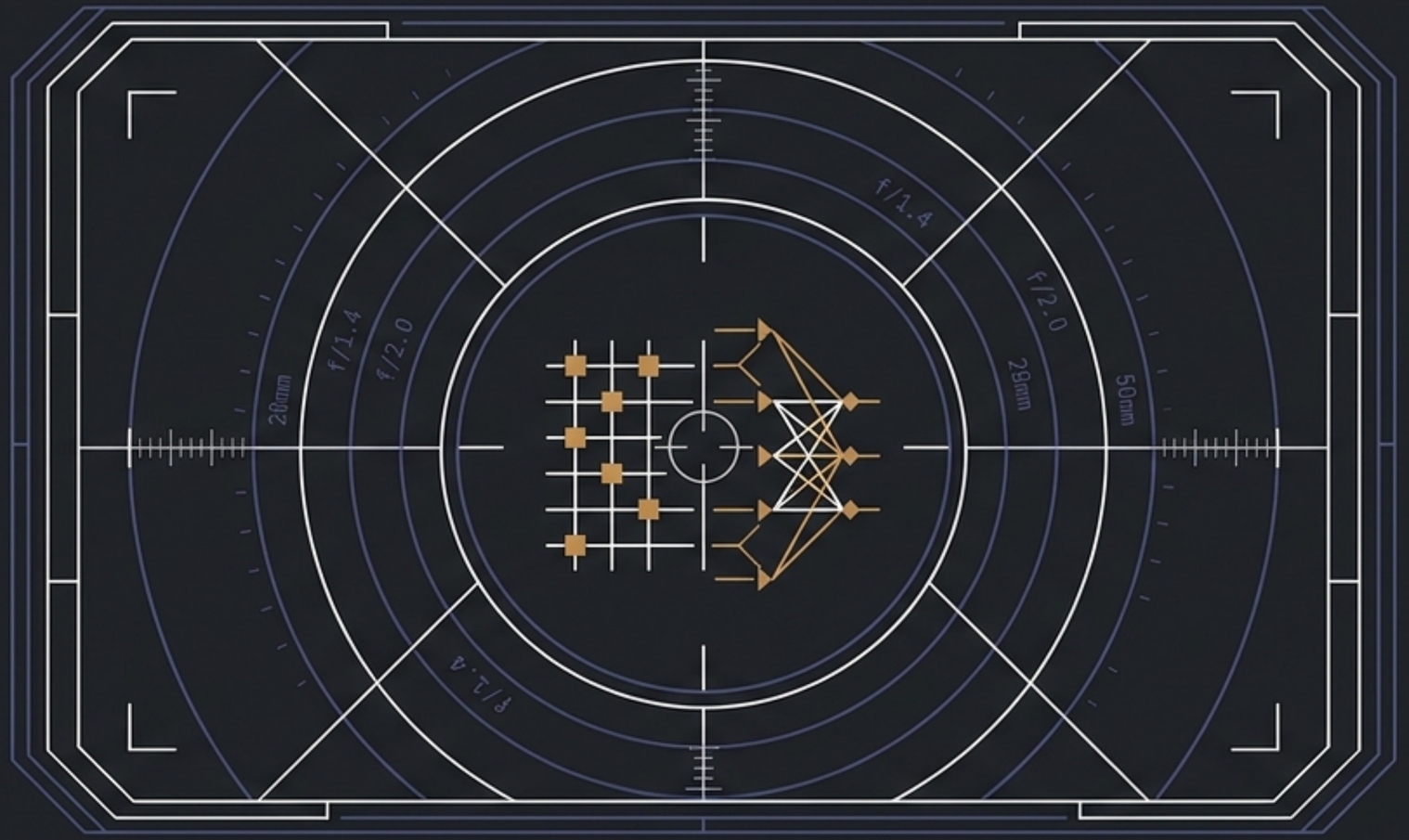


Kubona

See, observe, discover — then know.

An open-source, MCP-native quantum-computing and AI-enhanced science discovery platform.

Gemini Notebook

Observation is the founding act of of science.

1. Observe

Surface quantum phenomena systematically.

2. Measure

Run autonomous AI agent experiments.

```
AGENT_STATUS: ACTIVE
QUANTUM_EXPERIMENT: QBIT_B01A
DATA_FLOW: 5.4 TB/s
```

3. Record

Capture results via an immutable provenance ledger.

4. Understand

Synthesize organized, reproducible discoveries.

Chitonga (Tonga, Zambia)

ku- (infinitive) + **-bona** (see, observe, perceive)

inffinitive

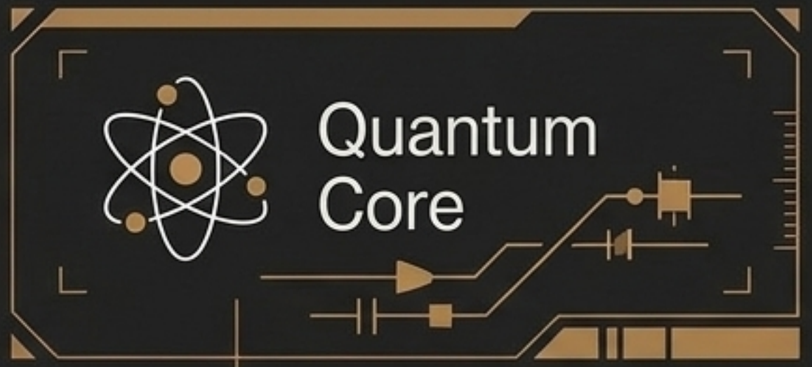
Gemini Notebook

The anatomy of an autonomous research laboratory.



Agentic Autonomy

- Multi-user lab environments
- User- and agent-created workflows
- Meta-level self-evolution loops



Quantum Core

- QIR/MLIR execution pipelines
- PySCF and Qiskit integration
- Built-in cost governance



Science Discovery

- Autonomous autoresearch engines
- Interactive experiment notebooks
- Organized, reproducible results

Gemini Notebook

Platform architecture & system topology.

Frontend (console/)

Sovereign Next.js UI on port 3000. Multi-user console and experiment studio.

`[TypeScript]`

Orchestration (core/)

The Go Orchestrator backend on port 8080. Federated MCP servers, provenance ledger, and core API.

`[Go]`

Execution (kubona/)

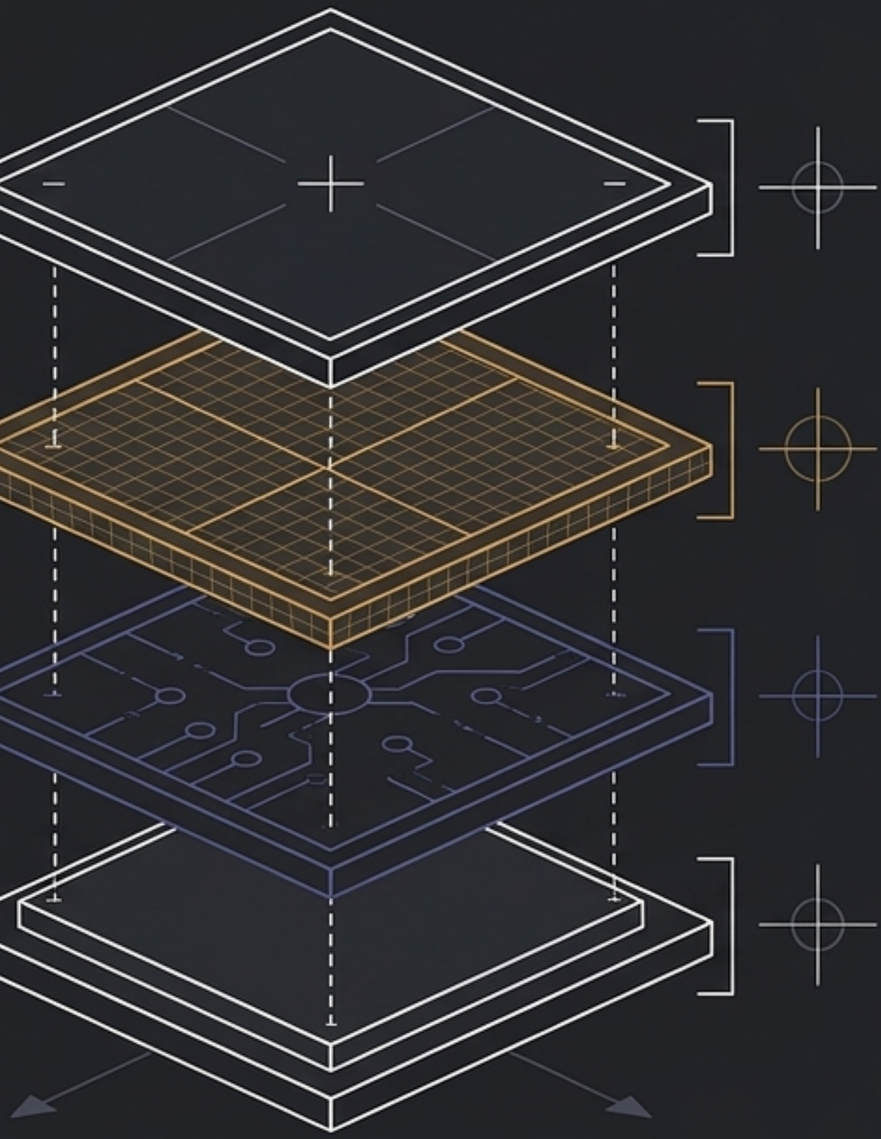
The Python runtime for agent execution and ML logic.

`[Python]`

Integration (bridge/)

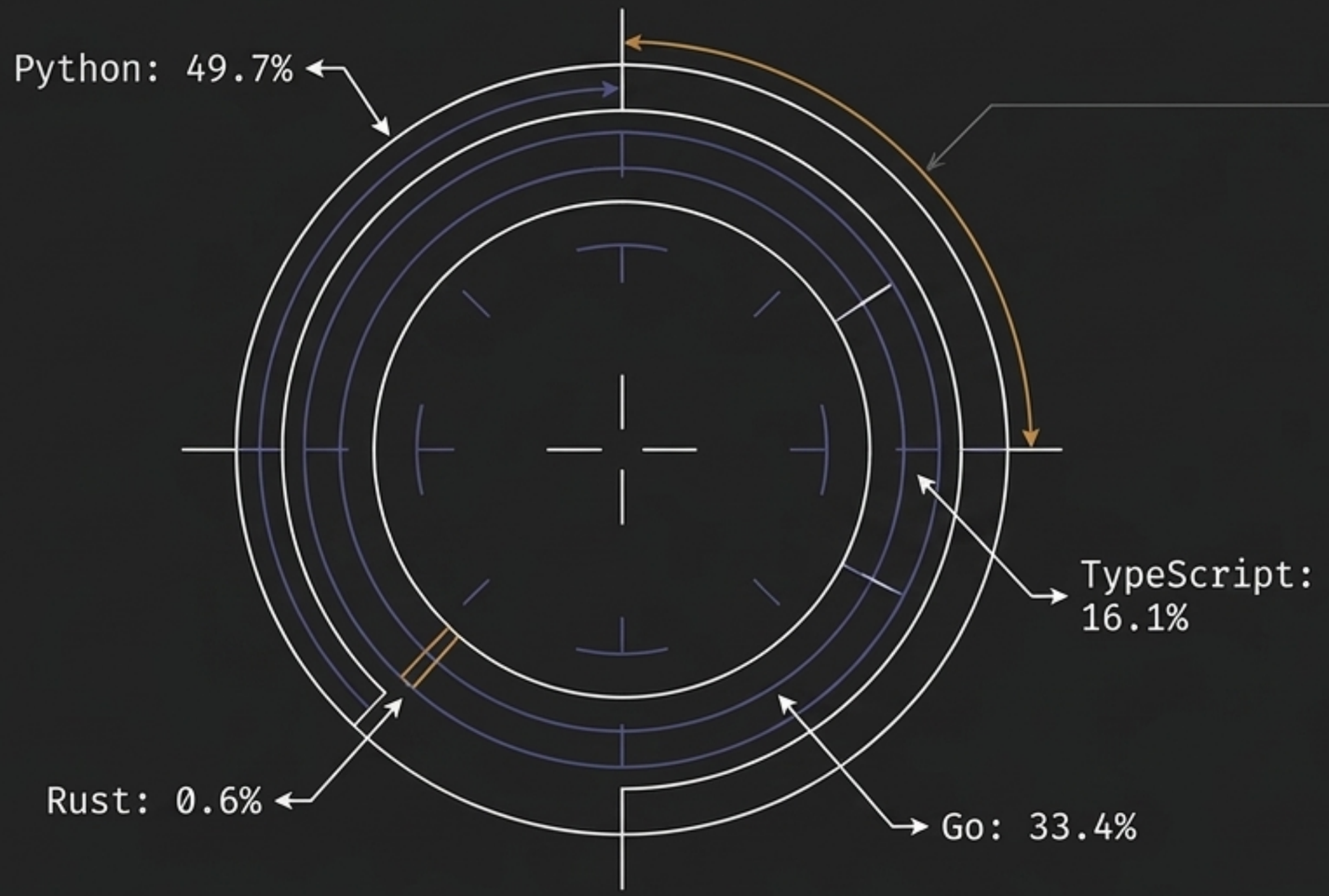
The PySCF/Qiskit bridge facilitating communication with quantum hardware and simulators.

`[Python/C++]`



Gemini Notebook

A meticulously structured, polyglot codebase.



wiki/

The living knowledge base. An Obsidian vault featuring 136 pages of concepts, indexes, and roadmaps (as of Oct 2026).

raw/

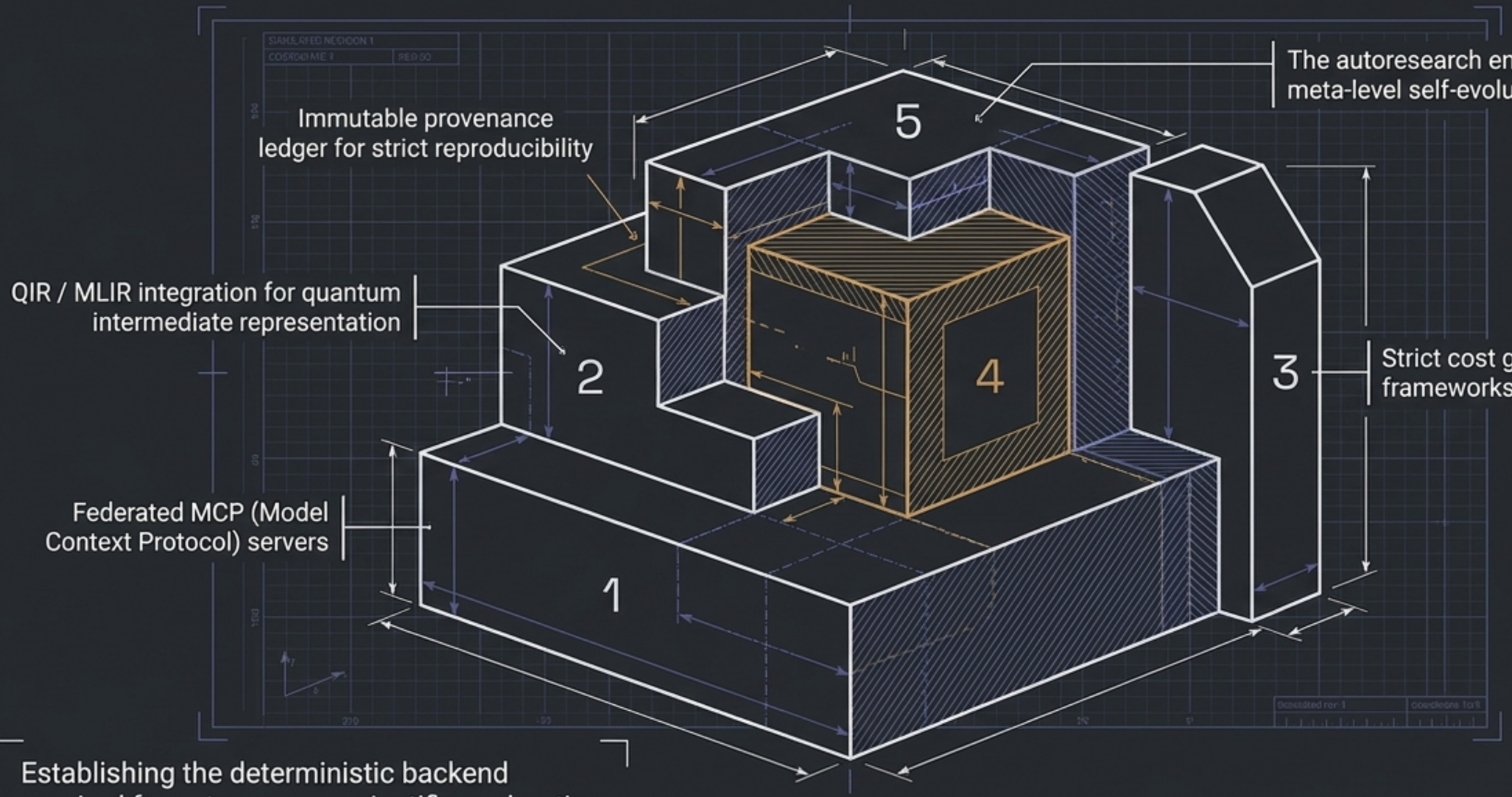
Immutable ingested source documents and implementation plans (read-only).

scripts/

CI/CD helpers, structural wiki linting, and automated code-sign re-runs.

Gemini Notebook

Phases 0–5: Forging the Quantum Core



Establishing the deterministic backend required for autonomous scientific exploration

Gemini Notebook

Phase 6: Science Synthesis & Discovery.



Translating quantum execution into actionable human knowledge.

Gemini Notebook

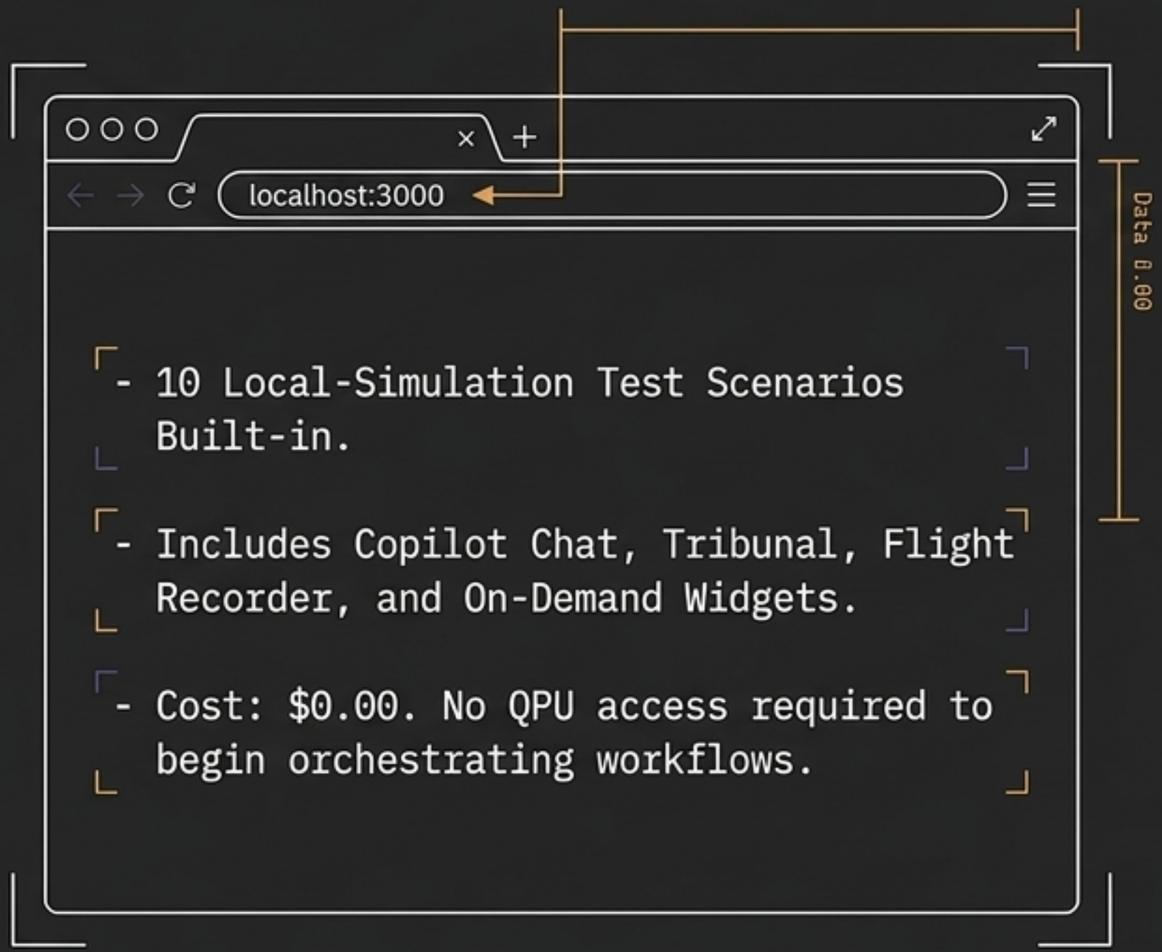
Zero-cost, zero-QPU local simulation

```
> go build -o /tmp/kubonad ./cmd/kubonad

> /tmp/kubonad -port 8080

> NEXT_PUBLIC_KUBONA_CONSOLE_TOKEN=dev-token
pnpm dev --port 3000
```

Data 50



Gemini Notebook

Rigor at every layer of the stack.

System Health STATUS: OK

Python Verification Suite

`.venv/bin/python -m pytest platform/tests/ -q`

961 passed, 3 skipped.
Full structural verification.

System Health STATUS: OK

Evaluator-Guided Circuit Co-Design

Advanced closed-vocabulary RPC rails and snippets collection ensuring code-generation validity.

System Health STATUS: OK

CI & Linting

Automated structural linting for all 136 wiki documents, enforcing maintainer operating rules (AGENTS.md).

DATA 1.02

Gemini Notebook

Discover — then know. Enter the lab.



Repo: mainza-ai/kubona
License: Apache-2.0 (Open Source)

- [ ] Read the 136-page living Obsidian knowledge base in /wiki.
- [ ] Spin up the local orchestrator and Next.js console.
- [ ] Run the 10 local-simulation test scenarios.

Gemini Notebook